WNDRVR
SUPPORT NETWORK

VXWORKS 7 VXBL DDR CONFIGURATION APP NOTE, SR0650

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12 June 2020

1. RECONFIGURING THE DDR REGISTER FOR USE WITH THE VXWORKS BOOT LOADER (VXBL)

Depending on your system configuration, you may need to adjust the method by which you configure the DDR registers in your system.

About This Task

Serial presence detect (SPD) is a hardware feature that enables you to automatically detect what memory is present in the system, and what timings are needed to access that memory. When a memory module supports SPD, certain parameters are required to be included in the lower 128 bytes of an EEPROM located on the memory module. This information includes timing parameters, manufacturer information, serial number, and other specific information related to the module. By accessing this data, you can automatically detect key information about the module which allows the system to correctly set certain parameters without user input.

In the current VxWorks 7 release, the VxWorks boot loader (VxBL) configures the DDR register using immediate integers (see the **ddrInit.c** file in any VxBL-enabled BSP). This configuration method reduces the code footprint and is more easily implemented compared to the SPD configuration method. However, the set of immediate integers described in the Supplement Guide for the BSP is specific to a particular DDR memory module. If the DDR memory module changes and you do not update the DDR configuration, VxBL can hang during the DDR controller initialization stage. This is often seen on Freescale boards between different board revisions. (That is, some newer revision boards can change to use another DDR memory module from a different vendor. This requires that you change the DDR configuration parameters in VxBL.)

Note: Currently, none of the ARM boards supported in VxWorks® 7 include a DDR memory module so this is not an issue for ARM systems.

To avoid this behavior, you can use U-Boot to retrieve the appropriate configuration parameters and you can then adjust your configuration as needed.

This application note describes how to change the DDR configuration parameters for a Freescale T2080QDS target board. Other Freescale boards can be configured in a similar manner.

Procedure

1. Dump the DDR parameters during the U-Boot boot stage.
    1. Enable the DEBUG macro in the **include/common.h** file from the U-Boot source tree.

    ```
    #define DEBUG
    #ifdef DEBUG
    #define _DEBUG  1
    #else
    #define _DEBUG  0
    #endif
    ```

    2. Rebuild and burn the U-Boot image.

    After power on, the memory control registers appear. The following is a partial example of the U-Boot dump:

    ```
    FSL Memory ctrl register computation
    FSLDDR: cs[0]_bnds = 0x0000007f
    FSLDDR: cs[0]_config = 0x80014302
    FSLDDR: cs[0]_config_2 = 0x00000000
    FSLDDR: cs[1]_bnds = 0x00000000
    ```


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```
FSLDDR: cs[1]_config = 0x00000000
FSLDDR: cs[1]_config_2 = 0x00000000
FSLDDR: timing_cfg_0 = 0x90110110
FSLDDR: timing_cfg_3 = 0x020a1100
FSLDDR: timing_cfg_1 = 0xe4ea3068
FSLDDR: timing_cfg_2 = 0x0051019b
FSLDDR: ddr_cdr1 = 0x80040000
FSLDDR: ddr_cdr2 = 0x00000001
FSLDDR: ddr_sdram_cfg = 0xe7040000
DDR: ddr_data_init = 0xdeadbeef
FSLDDR: ddr_sdram_cfg_2 = 0x24401110
FSLDDR: ddr_sdram_mode = 0x00441024
FSLDDR: ddr_sdram_mode_3 = 0x00001024
FSLDDR: ddr_sdram_mode_5 = 0x00001024
FSLDDR: ddr_sdram_mode_5 = 0x00001024
FSLDDR: ddr_sdram_mode_2 = 0x00a80000
FSLDDR: ddr_sdram_mode_4 = 0x00a80000
FSLDDR: ddr_sdram_mode_6 = 0x00a80000
FSLDDR: ddr_sdram_mode_8 = 0x00a80000
FSLDDR: ddr_sdram_interval = 0x10400100
FSLDDR: clk_cntl = 0x02000000
FSLDDR: timing_cfg_4 = 0x00000001
FSLDDR: timing_cfg_5 = 0x05401400
FSLDDR: zq_cntl = 0x89080600
FSLDDR: wrlvl_cntl = 0x8675f608
FSLDDR: wrlvl_cntl_2 = 0x090a0b0d
FSLDDR: wrlvl_cntl_3 = 0x0e0f110b
```

2. Replace the ddrCtrlInit( ) function in the VxBL.

   To set the DDR register value, use the values you obtained in the previous step.

```
LOCAL void ddrCtlrInit  (UINT32  regBase )
    {
    ddrIoWrite32(regBase + CS0_CONFIG            ,0x80014302);
    ddrIoWrite32(regBase + CS0_CONFIG_2          ,0x00000000);
    ddrIoWrite32(regBase + TIMING_CFG_0          ,0x90110110);
    ddrIoWrite32(regBase + TIMING_CFG_3          ,0x020a1100);
    ddrIoWrite32(regBase + TIMING_CFG_1          ,0xe4ea3068);
    ddrIoWrite32(regBase + TIMING_CFG_2          ,0x0051019b);
    ddrIoWrite32(regBase + DDRCDR_1,              0x80040000);
    ddrIoWrite32(regBase + DDRCDR_2,              0x00000001);
    ddrIoWrite32(regBase + DDR_DATA_INIT         ,0x00000000);
    ddrIoWrite32(regBase + DDR_SDRAM_CFG_2       ,0x24401110);

    ddrIoWrite32(regBase + DDR_SDRAM_MODE_CFG    ,0x00441024);
    ddrIoWrite32(regBase + DDRx_DDR_SDRAM_MODE_3 ,0x00001024);
    ddrIoWrite32(regBase + DDRx_DDR_SDRAM_MODE_5 ,0x00001024);
    ddrIoWrite32(regBase + DDRx_DDR_SDRAM_MODE_7 ,0x00001024);
    ddrIoWrite32(regBase + DDR_SDRAM_MODE_CFG_2  ,0x00a80000);
    ddrIoWrite32(regBase + DDRx_DDR_SDRAM_MODE_4 ,0x00a80000);
    ddrIoWrite32(regBase + DDRx_DDR_SDRAM_MODE_6 ,0x00a80000);
    ddrIoWrite32(regBase + DDRx_DDR_SDRAM_MODE_8 ,0x00a80000);
    ddrIoWrite32(regBase + DDR_SDRAM_INTERVAL    ,0x10400100);
    ddrIoWrite32(regBase + DDR_SDRAM_CLK_CTRL    ,0x02000000);
    ddrIoWrite32(regBase + TIMING_CFG_4          ,0x00000001);
    ddrIoWrite32(regBase + TIMING_CFG_5          ,0x05401400);
```


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```
    ddrIoWrite32(regBase + DDR_ZQ_CNTL           ,0x89080600);
    ddrIoWrite32(regBase + DDR_WRLVL_CNTL        ,0x8675F608);
    ddrIoWrite32(regBase + DDR_WRLVL_CNTL_2      ,0x090a0b0d);
    ddrIoWrite32(regBase + DDR_WRLVL_CNTL_3      ,0x0E0f110b);
    for (i = 0; i < 0x10000; i++);
    ddrIoWrite32(regBase + DDR_SDRAM_CFG         ,0xe7040000);
    }
```

Next Steps

You may also need to adjust the following:

- Use the **DDR_LAW** entry to adapt the **CS0_BNDS** value. (The entry is located in **vxbl/romInit.s**.)
- Set **DDR_SDRAM_CFG** after all other registers are set.

> Note: Bit 0 in the **DDR_SDRAM_CFG** register is the **MEM_EN** bit which indicates that the SDRAM interface logic is enabled. The **MEM_EN** bit must not be set until all other memory configuration parameters have been appropriately configured by the initialization code.

- Enable the **DDR_LAW** entry before enabling the DDR controller.
- If the DDR memory size changes, change the **LOCAL_MEM_SIZE** value in **vxbl.config** (located in the **vxbl** directory).

> Note: If you follow these instructions and you still see a failure in VxBL, check with the memory module vendor to ensure there are no additional errata available for the module.

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